METRIC

MIL-DTL-38999/20H
w/ AMENDMENT 1
11 August 2022
SUPERSEDING
MIL-DTL-38999/20H
27 July 2021

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, RECEPTACLE, THREADED, WALL MOUNTING FLANGE, REMOVABLE CRIMP CONTACTS, SERIES III, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

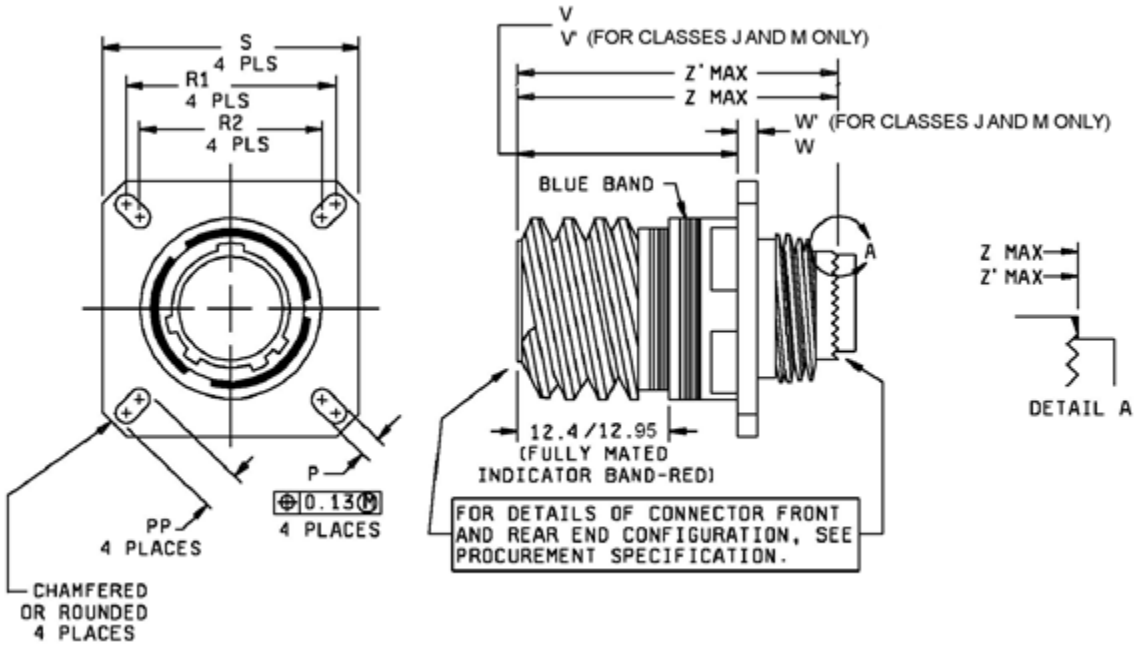


FIGURE 1. Receptacle.

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Shell size	Shell size code	M max, (see note 5)	P ± 0.20	PP ± 0.20	R1 (see note 3)	R2 (see note 4)	S ± 0.3
9	A	5.80 (.228)	3.25 (.128)	5.49 (.216)	18.26 (.719)	15.09 (.594)	23.8 (.937)
11	B		3.25 (.128)	4.93 (.194)	20.62 (.812)	18.26 (.719)	26.2 (1.031)
13	C		3.25 (.128)	4.93 (.194)	23.01 (.906)	20.62 (.812)	28.6 (1.126)
15	D		3.25 (.128)	4.39 (.173)	24.61 (.969)	23.01 (.906)	31.0 (1.220)
17	E		3.25 (.128)	4.93 (.194)	26.97 (1.062)	24.61 (.969)	33.3 (1.311)
19	F		3.25 (.128)	4.93 (.194)	29.36 (1.156)	26.97 (1.062)	36.5 (1.437)
21	G	5.00 (.197)	3.25 (.128)	4.93 (.194)	31.75 (1.250)	29.36(1.156)	39.7 (1.563)
23	H		3.91 (.154)	6.15 (.242)	34.93 (1.375)	31.75 (1.250)	42.9 (1.689)
25	J		3.91 (.154)	6.15 (.242)	38.10 (1.500)	34.93 (1.375)	46.0 (1.811)

Shell size	Shell size code	V + 0.00 - 1.25	V' + 1.4 - 0.0	W	W'	Z max	Z' max	External bending moment, minimum, classes J and M (N–m)
9	A	20.83	19.5	2.5/2.1	3.65/2.1	31.5 (1.240)	32.0 (1.260)	8.865
11	B	20.83	19.5	2.5/2.1	3.65/2.1	31.5 (1.240)	32.0 (1.260)	29.492
13	C	20.83	19.5	2.5/2.1	3.65/2.1	31.5 (1.240)	32.0 (1.260)	34.485
15	D	20.83	19.5	2.5/2.1	3.65/2.1	31.5 (1.240)	32.0 (1.260)	40.566
17	E	20.83	19.5	2.5/2.1	3.65/2.1	31.5 (1.240)	32.0 (1.260)	50.781
19	F	20.83	19.5	2.5/2.1	3.65/2.1	31.5 (1.240)	32.0 (1.260)	51.450
21	G	20.07	18.7	3.2/2.1	4.35/2.1	31.5 (1.240)	32.0 (1.260)	80.941
23	H	20.07	18.7	3.2/2.1	4.35/2.1	31.5 (1.240)	32.0 (1.260)	69.810
25	J	20.07	18.7	3.2/2.1	4.35/2.1	31.5 (1.240)	32.0 (1.260)	91.310

NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are given for information only.
3. Front or rear panel mounting.
4. Front panel mounting only.
5. Front or rear panel thickness. To ensure full mating of the plug or the backshell, it is recommended that the front or rear mounting panel thickness not exceed the M values specified in the table.
6. The specified red band location is measured from the front edge of the shell to the back edge of the red band.
7. Dimensions V', W', and Z' are for classes J and M, only. Dimensions V, W, and Z are for all other classes.

FIGURE 1. Receptacle – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1. Interface dimensions shall be in accordance with MIL-DTL-38999.

This connector mates with MIL-DTL-38999/26.

For insert arrangements: See MIL-STD-1560.

Connector accessories: See SAE-AS85049.

Part or Identifying Number (PIN) example:

D38999/	20	W	A	35	P	N

D38999/	DoD number prefix
20	Specification sheet number
W	Class
A	Shell size code
35	Insert arrangement
P	Contact style
N	Polarizing positions (N is required for normal position)

Restrictions: Class F is not for Navy use and is inactive for Air Force new design use.

NOTE: The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification.

Qualification required: The activity responsible for the qualified product list for this specification sheet is DLA Land and Maritime-VQ, 3990 East Broad Street, Columbus, Ohio, 43218-3990.

Note: Bending moment torque (class J and M). Bending moment requirements were determined through finite element analysis (FEA) and testing parts to yield strength. Material yield strength was characterized by a normal distribution with $\mu_{material,yield} = 275.79MPa$ and $\sigma_{material,yield} = 13.78\ MPa$. A yield stress divided by yield torque ratio (α_{yield}) and a real world torque variation (σ_r) were determined and given in Table I. Requirements ($\tau_{requirement}$) were set at 3 standard deviations from the mean as given below:

$$\tau_{requirement} = \frac{\mu_{material\ ,yield}}{\alpha_{yield}} - 3 * \sqrt{\frac{{\sigma_{material\ ,yield}}^2}{{\alpha_{yield}}^2} + {\sigma_r}^2}$$

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TABLE I. Bending moment torque parameters

Shell Size	$\alpha_{yield}\left(\frac{MPa}{N*m}\right)$	$\sigma_r(N*m)$
9	25.362	0.391
11	7.540	1.494
13	6.610	1.213
15	4.620	5.636
17	4.386	2.527
19	4.165	3.642
21	2.686	5.116
23	2.669	9.902
25	2.270	8.022

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-STD-1560
MIL-DTL-38999/26
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - AS
Air Force – 85
DLA - CC

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5935-2022-178)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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